CreatorGuide Agent

A grounded, tool-using assistant for YouTube creator policies - capstone report

Poonnada Sripoonsombat · LLM application capstone · built 04 November 2026

1. Problem and users

YouTube creators constantly ask rule questions: how long a strike lasts, whether a strike can still be appealed, what a counter notification is, how many subscribers and watch hours the Partner Program needs. The authoritative answers live in dozens of Help articles, and the questions are often multi-step ("I have 650 subscribers and 3,200 watch hours - how many more do I need?"). A model answering from memory is risky here: thresholds change, and a wrong answer about strikes or monetization has real cost for a creator.

CreatorGuide Agent answers such questions only from a frozen corpus of 62 English YouTube Help articles (policy, enforcement, monetization; snapshot of 5 and 12 Oct 2026). Every answer is a validated JSON object with citations to the passages the model actually saw, and the system declines when the articles do not cover the question. It is for creators and for support staff who need a quick, checkable answer.

2. Design

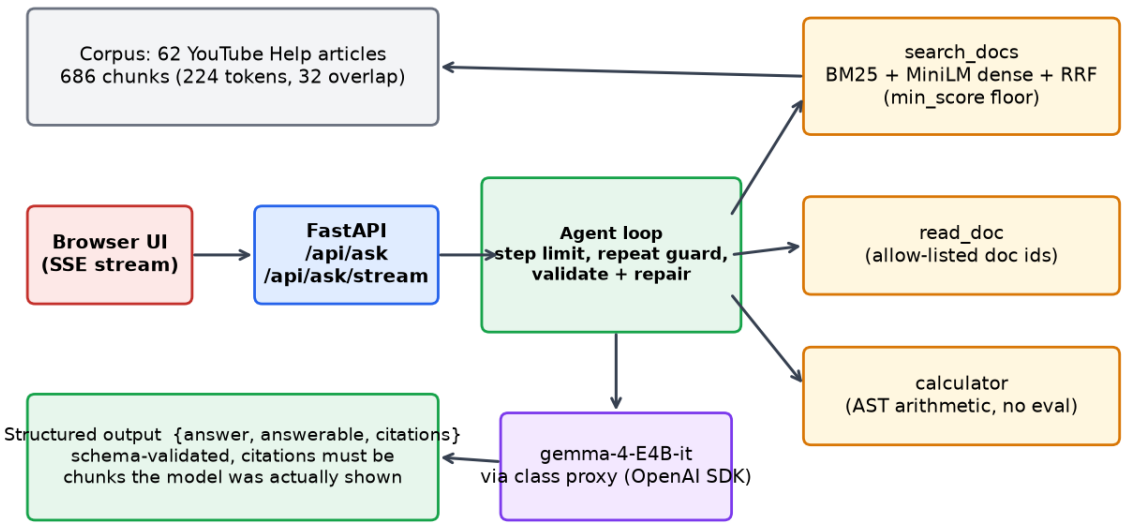


Figure 1. Architecture. The agent is the only component that talks to the model; tools never run anything the allow-list does not name.

Course technique	Where it is used	Why this choice
Structured output + schema validation	Final answer {answer, answerable, citations}; tool arguments	Small models drift from format. Validation rejects wrong types, unknown keys and citations the model was never shown, then sends the error back for up to 2 repairs; otherwise the system abstains.
Retrieval / RAG with citations	search_docs: BM25 + MiniLM dense, fused with Reciprocal Rank Fusion	Policy text mixes exact terms ("90-day", "YPP") with paraphrases. BM25 catches the former, dense the latter; RRF fuses ranks, not incomparable scores.
Retrieval evaluation	eval/: Hit@k, Recall@k, MRR on labelled documents; min_score and chunk-size sweeps	Measures retrieval separately from generation, so a bad answer can be traced to its cause.
Agent loop with tools (agentic)	search_docs, read_doc, calculator; max 6 tool rounds; repeat-call guard	Multi-step questions need search + arithmetic. A deterministic calculator removes small-model arithmetic errors.
Served interface	FastAPI JSON endpoint and an SSE stream rendered by a small web page	Streaming each tool call makes the agent's reasoning visible, which is also the demo.

Safety and robustness. API key and base URL come only from the environment (the OpenAI SDK reads them). Tool names are allow-listed and arguments are validated (types, lengths, ranges, unknown keys rejected). The calculator parses an AST and permits only numbers and arithmetic operators. Retrieved text and tool results are framed as untrusted quoted data in every prompt. Temperature is 0 for everything that is evaluated. Tool results are capped at 5,000 characters and always remain valid JSON.

3. Evaluation design

Test set. 30 questions written for this project: 20 single-fact, 5 multi-step (three need the calculator, two need facts from two different articles), and 5 unanswerable (off-corpus, including one adversarial case where "CPM" means Commerce Product Module in the corpus). Every answerable question has gold document ids, a verbatim evidence quote from the article (checked automatically by `eval/check_questions.py`), and key-fact groups: each group lists acceptable variants of one required fact (e.g. 350 and 800 for question m01).

Systems compared. *no_retrieval* (model alone, baseline 1); *rag_bm25* (naive default: BM25 without stopword filter, top-3, no relevance floor, single shot - baseline 2); *rag_ours* (single shot with hybrid retrieval, stopword filter, top-5, calibrated floor); *agent* (tool-calling loop on the same retriever).

Metrics. Retrieval (document level, answerable questions): Hit@1/3/5, Recall@3/5, MRR@10. Answers (deterministic, no LLM judge): *accuracy* = every key-fact group matched and the system chose to answer (for unanswerable questions: it abstained); *grounded* = correct and cites a gold document; *key-fact recall*; *abstention accuracy* on unanswerable questions; *false-abstain rate*; cost (LLM calls, tokens). Differences between systems are reported with a paired bootstrap 95% interval (10,000 resamples, seed 42) because with 30 questions a single question moves accuracy by 3.3 points.

4. Results

4.1 Retrieval

System	Hit@1	Hit@3	Hit@5	Recall@3	Recall@5	MRR@10
bm25	0.880	1.000	1.000	0.980	0.980	0.933
bm25_stopwords	0.920	1.000	1.000	0.960	0.960	0.953
dense	0.800	0.960	0.960	0.900	0.940	0.885
hybrid	0.920	1.000	1.000	0.960	0.980	0.960
hybrid_stopwords	0.880	1.000	1.000	0.960	0.980	0.940

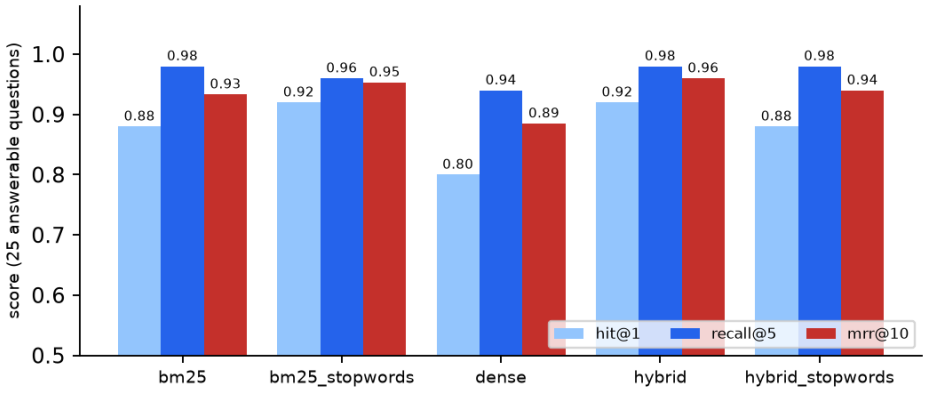


Table 1 / Figure 2. Retrieval on 25 answerable questions, 224-token chunks, 62 documents.

bm25_stopwords_vs_bm25_mrr@10: mean difference +0.020, 95% CI [-0.060, +0.093] - the interval includes 0, so this is not a reliable gain.

dense_vs_bm25_mrr@10: mean difference -0.048, 95% CI [-0.178, +0.078] - the interval includes 0, so this is not a reliable gain.

hybrid_vs_bm25_mrr@10: mean difference +0.027, 95% CI [-0.060, +0.120] - the interval includes 0, so this is not a reliable gain.

hybrid_stopwords_vs_bm25_mrr@10: mean difference +0.007, 95% CI [-0.093, +0.107] - the interval includes 0, so this is not a reliable gain.

4.2 Measured improvements over naive defaults

Stopword filter for BM25. Naive BM25 lets question words ("how long do I have to") outweigh content words. Removing a short list of function and question words (keeping "not", "no", "under", "at", "least", which change policy meaning) moves Hit@1 0.880 to 0.920 and MRR@10 0.933 to 0.953. The paired interval below shows whether this difference is reliable.

Relevance floor (min_score), chosen from data. For hybrid the gate signal is `dense_score` of the best top-5 hit. Without a floor the system rejects 0% of unanswerable questions. The chosen operating point, min_score = 0.5233, keeps 100% of answerable questions and rejects 100% of unanswerable ones (constraint: keep at least 90% of answerable). Only 5 unanswerable questions exist, so the estimate is rough.

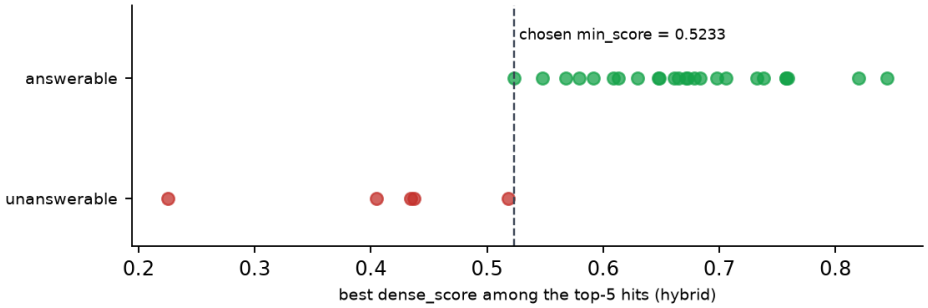


Figure 3. Gate score of every question; unanswerable questions are not cleanly separable by score alone.

Chunking (BM25 + stopwords)	Chunks	Hit@3	Recall@5	MRR@10
committed_224_token_index	686	1.000	0.960	0.953
words-60	1650	1.000	0.980	0.953
words-100	1035	1.000	0.980	0.973
words-150	726	1.000	0.980	0.960
words-250	554	1.000	0.980	0.953
words-400	473	1.000	0.980	0.953
sections	436	1.000	0.980	0.953

Table 2. Chunk-size sweep (word-based chunks, offline). MRR@10 ranges only from 0.953 to 0.973 across sizes: on 25 questions this is within noise, so I keep the 224-token index and do **not** claim a chunk-size improvement.

4.3 Answer quality (LLM)

System	Accuracy (30)	Single (20)	Multi (5)	Abstain ok (5)	Grounded	False abstain	LLM calls	Tokens
no_retrieval	30%	15%	40%	80%	0%	32%	1.0	224
rag_bm25	70%	70%	40%	100%	60%	20%	1.0	873
rag_ours	93%	100%	60%	100%	92%	0%	0.8	1023
agent	90%	95%	60%	100%	84%	8%	2.7	2809

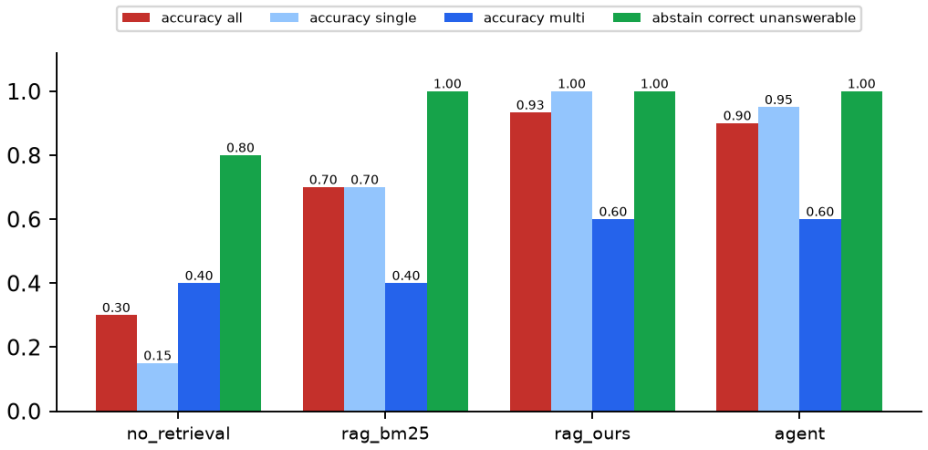


Table 3 / Figure 4. Answer accuracy with gemma-4-E4B-it, temperature 0.0, retrieval = hybrid.

rag_bm25_vs_no_retrieval_accuracy_all: +0.400 (95% CI [+0.167, +0.600]) - excludes 0.

rag_ours_vs_no_retrieval_accuracy_all: +0.633 (95% CI [+0.467, +0.800]) - excludes 0.

agent_vs_no_retrieval_accuracy_all: +0.600 (95% CI [+0.433, +0.767]) - excludes 0.

no_retrieval_vs_rag_bm25_accuracy_all: -0.400 (95% CI [-0.600, -0.167]) - excludes 0.

rag_ours_vs_rag_bm25_accuracy_all: +0.233 (95% CI [+0.100, +0.400]) - excludes 0.

agent_vs_rag_bm25_accuracy_all: +0.200 (95% CI [+0.000, +0.400]) - includes 0.

agent_vs_rag_ours_accuracy_all: -0.033 (95% CI [-0.133, +0.067]) - includes 0.

5. Analysis and failure cases

Systems ranked by overall accuracy: rag_ours 93%, agent 90%, rag_bm25 70%, no_retrieval 30%. The agent answers 1 fewer question(s) correctly than single-shot RAG overall, so tool use did not improve accuracy here; on the 5 multi-step questions the two tie, so the calculator tool shows no accuracy benefit on this set. The agent uses 2.7x the tokens and 2.7 vs 0.8 LLM calls per question. With 30 questions each question is 3.3 points, so read the intervals above before claiming that one system is better.

Our single-shot RAG configuration (hybrid retrieval, stopword filter, top-5, relevance floor) vs the naive default (BM25, top-3, no floor): +23.3 points (95% CI [+10.0, +40.0]); the interval excludes 0. This compares a bundle of changes, so it does not show which of them matters; the retrieval-only comparison in 4.1 shows small differences.

Failure 1 - agent / s02. Q: What subscriber and watch-hour minimums let creators start earning through fan funding, Creator Partnerships or YouTube Shopping?
Expected: 500; 3,000/3000
Got (final, answerable=False): The provided articles state that to earn revenue from fan funding features, a channel must be in the YouTube Partner Program and meet other requirements, but they do not specify the exact subscriber and watch-hour minimums for fan funding features like Channel

Failure 2 - agent / m02. Q: I have 420 subscribers and 2,700 qualified watch hours in the past year. How many more of each do I need to start earning through fan funding, Creator Partnerships or YouTube Shopping?
Expected: 80; 300
Got (final, answerable=False): The provided YouTube Help articles do not specify the exact subscriber and watch hour requirements needed to start earning through fan funding, Creator Partnerships, or YouTube Shopping. The articles mention that to turn on fan funding features, a channel must

Failure 3 - agent / m05. Q: What can happen if my channel gets 3 strikes in the same 90 days, and can I just make a new channel afterwards?
Expected: removed/terminat/permanent; prohibited/cannot/can't/not allowed/may not/unable/not able/won't
Got (final, answerable=True): If you receive 3 strikes within 90 days, your channel may be terminated. The articles mention that for repeated violations of the Community Guidelines or Terms of Service, YouTube may terminate your channel or account. However, the provided articles do not sta

Failure 4 - rag_ours / m03. Q: I got a Community Guidelines strike 5 months ago. Can I still appeal it, and is the strike still active?
Expected: 6 months/six months/still; expire/90 days/no longer active/not active
Got (final, answerable=True): You can appeal for 6 months after the warning or strike was issued [yt-185111:0001]. Since 5 months have passed, you are within the appeal window. If you appeal, you may be told that the strike will stay if the content was found to be in violation of the Commu

Retrieval failure analysis (measured). Naive BM25 ranks a gold document first for all but 3 of 25 questions. Misses: s06 ("How long do I have to appeal a Community Guidelines strike?", top-3: yt-2802032 yt-185111 yt-2807684; fixed by the stopword filter: yes); s07 ("How long after my content was removed can I submit an appeal?", top-3: yt-1727191 yt-2807684 yt-185111; fixed by the stopword filter: no); s15 ("Can I start a new channel after mine was terminated?", top-3: yt-1727191 yt-2802168 yt-12504220; fixed by the stopword filter: yes). The pattern is lexical: chatty question words and generic terms ("strike", "channel") match many articles, so a document about a neighbouring topic can win.

Additional observations. After the first full run I reviewed every failure. Two questions (s02, m02) used the phrase "early-access tier", which the articles never use, so systems that declined to answer were arguably right; I reworded them using the corpus's own terms. The key facts of s05 were widened because "two additional weeks" is a correct answer, and the variant "no" in s15 and m05 was replaced because it also matched "there is no restriction". All 30 questions were re-run for all four systems; first-run results are kept in eval/results_v1. Because these revisions were made after seeing results on the same set, the reported figures are optimistic compared with a fresh held-out set.

6. Limitations and next steps

Small test set. 30 questions (25 answerable, 5 unanswerable) written by one person: intervals are wide and the min_score estimate is rough. **Key-fact matching** is deterministic and strict about wording variants I listed; a correct answer phrased differently can be marked wrong, and a wrong answer that happens to contain the numbers can be marked right (no LLM judge was used, because a 4B model judging itself is not trustworthy). **Corpus freshness.** The articles are a 5-12 Oct 2026 snapshot. Several already announce Partner Program changes effective 1 Feb 2027 (for example higher entry thresholds for new creators), so the gold answers of the threshold questions are valid only until 31 Jan 2027 and the model can be confused by the future-dated text. **Model.** gemma-4-E4B-it is small: it can ignore tools or format; the validate-repair-abstain loop contains the damage but costs extra calls. **Security.** Prompt injection from retrieved text is mitigated by framing and by tool allow-listing, not eliminated; there is no authentication or rate limiting on the API.

Next steps: grow the set to 100 questions with a second annotator; add a cross-encoder reranker and measure it against hybrid; add an LLM-judge faithfulness score validated against human labels; expose the three tools through an MCP server; add Thai-language questions.

Reproducibility

All numbers come from `eval/results/`, regenerated by `make eval-offline` and `make eval-answers`. Question-set sha256 c3ca916355939686..., corpus sha256 71ef16638f0b6f68.... Dense retrieval uses the pinned all-MiniLM-L6-v2 revision stored in data/index/manifest.json.